

What is the obesity status of Latin American women and which factors are associated with their risk?

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Executive Summary

Our project examines the obesity status of Latin American women and the factors that are associated with their risk using a data sample of 1043 observations extracted from Kaggle Obesity Prediction dataset. The data contains information on dietary habits, physical activity and physical characteristics for women from Colombia, Mexico and Peru.

To explore potential determinants of obesity, we classified individuals into two categories based on their BMI level: non-obese (levels 0-3) and obese (levels 4-7).

We focus on two explanatory variables which could be significant to explore: **age** and physical activity level (**pa1**).

Our objective is to first compare the mean BMI level of our sample to worldwide reference value to assess how Latin American women situate themselves relative to global trends. Then to determine whether our two explanatory variables differ significantly between the two BMI groups.

All statistical analyses rely on inferential methods, notably hypothesis testing for mean differences. Our findings reveal three key results:

1. The average BMI level in our sample (4.32) exceeds the global average of 3.0
2. Obese women are significantly older than non-obese women with a mean difference of 3.80
3. Obese women have significantly lower physical activity levels than non-obese women, suggesting that sedentary behavior plays a meaningful role in BMI disparities.

Taken together, these results indicate that higher age and lower physical activity are all associated with obesity among Latin American women. Overall, this project highlights the importance of addressing multiple lifestyle-related factors simultaneously when designing public health interventions in the region.

1. Introduction

In 2022, the World Health Organization (WHO) reported that one in eight adults worldwide was living with obesity. In Latin America, obesity represents a major public health concern, partly due to changes in dietary habits and increasingly sedentary lifestyles. Research also shows that women are disproportionately affected by obesity in many regions of the world. According to the Pan American Health Organization (PAHO), adult women in Latin America consistently exhibit higher obesity rates than men, making them a particularly relevant demographic for focused analysis.

In this project, we aim to identify which individual characteristics available in our dataset may be statistically associated with a higher risk of obesity among Latin America women. Our objective is not to establish causality but to examine whether certain personal and lifestyle related variables show significant differences between obese and non-obese individuals within our sample.

We use the Obesity Prediction dataset from Kaggle, which provides information on diet, physical activity and physical characteristics of individuals from Colombia, Mexico and Peru. From this raw dataset, we constructed a subset containing only women that gave a sample of 1043 observations, which forms the basis of our analysis. Our dependent variable is the **imc_level**, calculated using weight and height. For part of our analysis, BMI levels were grouped into two categories: non-obese (levels 0 to 3) and obese (levels 4 to 7).

We consider two explanatory variables: **age** and **physical activity level**. Using statistical inference, and particularly hypothesis testing, we seek to determine whether these variables are associated with significant differences in BMI levels within our sample of Latin American women and help to understand their prevalence of obesity.

2. Data

2.1. Data gathering

We have chosen to focus our study on the risk of obesity among women in Latin America. The data are drawn from the “Obesity Prediction Dataset” available on Kaggle, which compiles information on dietary habits, physical characteristics, and activity levels on individuals from Colombia, Mexico, and Peru.

We first imported the Kaggle dataset into Excel. We created a new dataset by retaining only the observations corresponding to women. Based on this filtered dataset, we extracted a sample of 1043 individuals, which forms the database used for our inferential analysis. This dataset is therefore a sample, as it represents only a small part of the women population in Latin America.

The dependent variable of our study is **imc_level**, a categorical indicator derived from the Body Mass Index (BMI). Since **imc_level** was not provided directly, we calculated it using weight and height according to the standard formula
$$\text{imc_level} = \frac{\text{weight}}{\text{height}^2}$$

We then classified everyone into one of the eight BMI categories based on the reference classification published by “Medicine Anti-age” (2024)¹ (Appendix C). For our hypothesis testing, we grouped these eight categories into two levels:

- Non-obese (0): underweight, normal weight and overweight
- Obese (1): all obesity categories

This binary version of **imc_level** enables us to perform two-group comparisons in the hypothesis-testing section.

To analyze potential differences between the two groups, we selected two explanatory variables from the Kaggle dataset:

- Age: continuous variable expressed in year, which we renamed **age**
- FAF: continuous variable representing the frequency of physical activity, which we renamed **pal** (Physical Activity Level)

The variable **pal** is a measure coded from 0 to 3 (0 = no activity,]0,1] = low,]1,2] = moderate,]2,3] = frequent). These renaming were made to improve clarity and facilitate interpretation in the analysis (Appendix D).

2.2 Descriptive overview

Table 1: Significant data for our variables

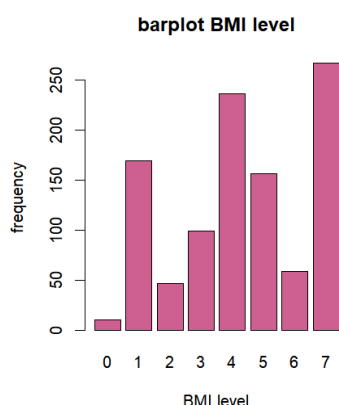
Summary Var	Minimum	1st Quantile	Median	Mean	3rd Quantile	Maximum
Age	15.00	19.00	22.00	23.65	26.00	61.00
PAL	0.0000	0.0000	0.7421	0.8471	1.5073	3.0000
IMC_level	0.000	3.000	4.000	4.321	7.000	7.000
Obesity_level	0.000	0.000	1.000	0.6884	1.000	1.000

The descriptive table indicates that the sample is overall young with a low level of physical activity. The BMI values are mainly high and approximately 69% of the participants are classified as obese. We will continue to comment on these values in the following part.

¹ Medicine Anti-Age. (s2025.). Indice de masse corporelle IMC. <https://www.medecine-anti-age.com/sport/imc>

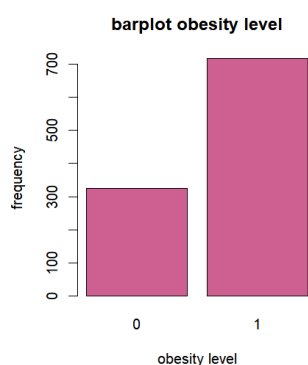
2.3 Description of the Dependent Variable

Figure 1: Main variable diagram -BMI level



The bar chart illustrates the distribution of BMI categories from 0 to 7. It shows that most of the observations fall within categories 4, 5 and 7, corresponding to elevated BMI levels ranging from overweight to moderate obesity. The distribution is strongly right skewed, indicating that most women in the sample have a BMI exceeding the normal weight of category 3. The median for this variable is 4.

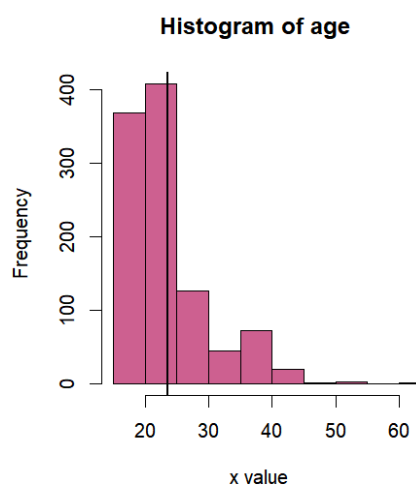
Figure 2: Main variable diagram -obesity status for the two subgroups (obesity_level)



This bar chart represents the binary obesity variable derived from BMI. Category 0 corresponds to non-obese women while category 1 represents obese women. The distribution is highly imbalanced: approximately 70% of the women belong to the “obese” group. The “non-obese” category remains in the minority, accounting for about 30% of the sample.

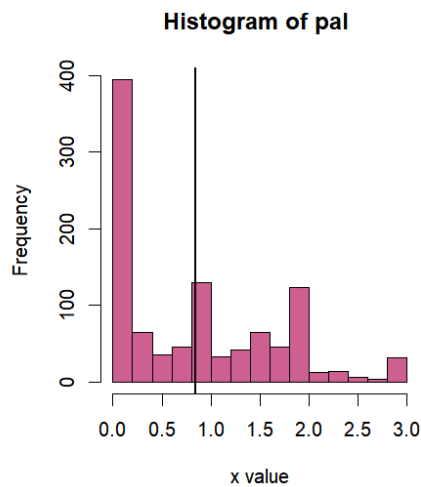
2.4 Presentation of associated explanatory variables

Figure 3: Age histogram



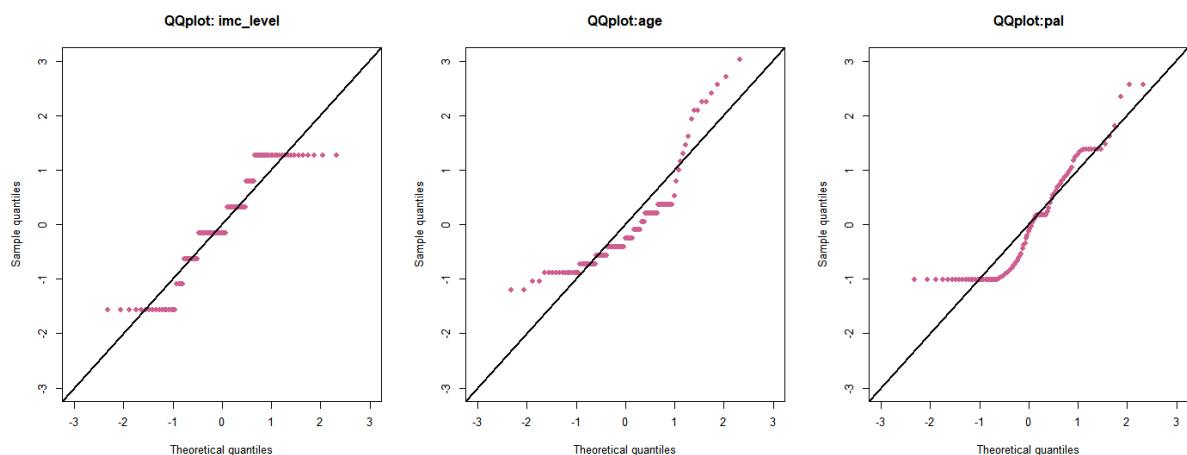
The histogram illustrates a distribution strongly skewed to the right. Most women in the sample are between 18 and 25 years old, with a notable concentration around their twenties. Older age groups are significantly less represented. The vertical line represents the mean age, which is 23.65 years, slightly higher than the median (22 years).

Figure 4: pal Histogram



The histogram for **pal** (Physical Activity Level) reveals a highly imbalanced distribution with a very strong concentration of participants around values close to 0, corresponding to a sedentary lifestyle. The median is 0.7421 which is lower than the mean of 0.8471. It shows that a few higher values slightly pull the distribution to the right. **pal** can therefore be considered a variable with low dispersion with most observations concentrated in the lower range.

Figure 5: QQ plots for the variables

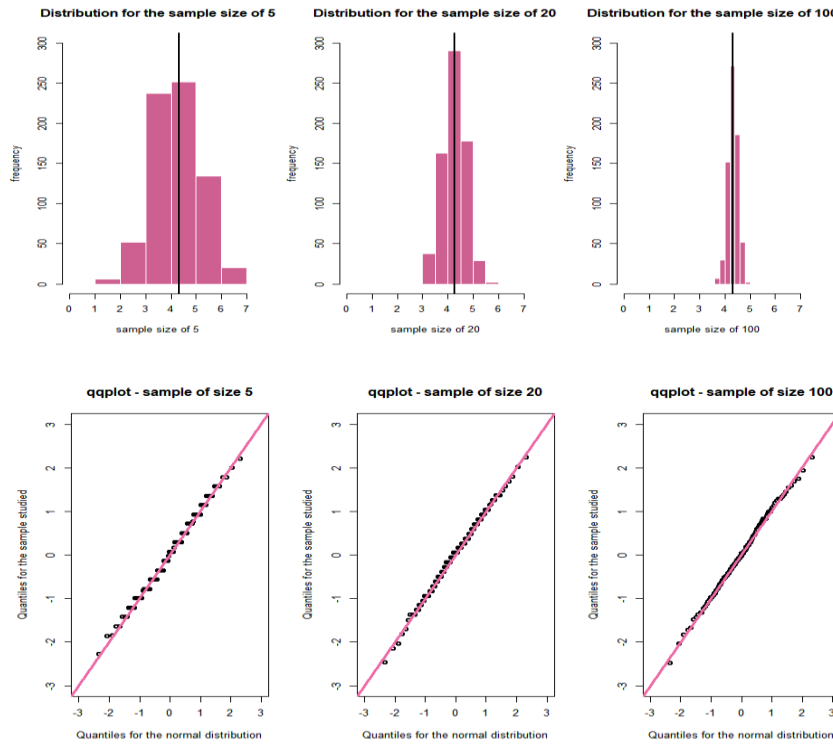


- The QQ plot for the **imc_level** has horizontally aligned groups of points, corresponding to a categorical variable ordered into 8 classes (from 0 to 7). It is not possible to model this type of variable using a normal distribution.
- For **age**, the plot shows that most points do not follow the reference line, particularly at the extremes. This suggests that age is not normally distributed.
- The QQ plot for **pal** shows that the value partly follows the normal distribution. Only the extreme values of this variable do not follow it.

2.5. The central limit theorem

For the analysis we will conduct forward, we will use the hypothesis test which is based on the fact that our data may approximate the normal model even though we saw that our main variable does not follow the normal distribution (Appendix A example of a perfect normal distribution).

Figure 6: Central limit theorem (CLT)



The figure illustrates the sampling distributions (700 samples) of the sample mean for the variable `imc_level` across different sample sizes: $n = 5$, 20 and 100. It shows the Central Limit Theorem, which states that the sampling distribution of the sample mean becomes increasingly normal and more tightly concentrated around the true population mean as n increase. For $n = 5$, the distribution is wide and irregular in shape. For $n = 20$, it becomes more symmetric and narrower. And then for $n = 100$, the effect of the Central Limit Theorem is clearly visible. The distribution is narrow, centered around the average and nearly normal.

3. Methodology

The analysis will primarily use statistical inferential methods. Based on our previous examination of the sample, particularly its relationship to the normal model, supported by the Central Limit Theorem, we can reliably use hypothesis testing as our primary method for obtaining results.

Our analysis will be divided into three sections, each corresponding to a hypothesis that will be tested. Below are the three hypotheses, each preceded by a brief justification.

Every geographic location has its particularities, and they differ by their culture, education, and behavior. All these factors can change the situation of a specific country compared to the rest of the world. So, we will examine if our sample aligns with the worldwide average BMI score

for women in 2016 ², which is 24.4 (corresponds to level 3 on our scale for BMI which corresponds to a Normal Weight). This will allow us to notice if this population is having a prevalence of obesity.

$H_0: \mu_1 = 3$ (μ_1 : mean BMI level of women in Latin America,)

$H_1: \mu_1 > 3$ (the mean BMI level is higher than the worldwide average)

We perform a right-tailed test on the mean.

For the last two hypotheses, we compare the two groups defined by the **obesity_level** based on the values of explanatory variables:

We assert that age is a factor potentially associated with obesity risk. Younger individuals, due to the increased sedentary lifestyle of our era, may be more prone to obesity. Also, with advancing age, metabolism slows down, which could promote weight gain in older individuals. Based on these observations, we establish the following hypothesis test:

$H_0: \mu_1 = \mu_2$ (μ_1 : mean **age** of obese women; μ_2 : mean **age** of non-obese women)

$H_1: \mu_1 \neq \mu_2$ (the mean **age** differs between the 2 groups)

We perform a two-tailed test on the mean.

Physical activity helps burn calories, which reduces weight gain. Regular exercise may help prevent obesity. So, we will test whether obese women have a lower average frequency of physical activity than the non-obese group.

$H_0: \mu_1 = \mu_2$ (μ_1 = mean physical activity frequency of obese women; μ_2 = mean physical activity frequency of non-obese women)

$H_1: \mu_1 < \mu_2$ (obese women have a lower mean frequency)

We perform a left-tailed test on the mean.

To facilitate the calculus of the two last hypotheses, we will express them in this form

$$H_0: \mu_1 - \mu_2 = 0$$

To determine if we reject the null hypothesis (H_0) in each case, we will calculate the p-value and compare it to the significance level α which we set at 5%. This value is a standard convention that allows us to identify meaningful evidence against H_0 without being overly sensitive and risking false positives (Type I error). If the calculated p-value is below this threshold, we reject H_0 , otherwise we retain it. So given the statistical approach used, the results are subject to a 5% margin of error regarding the rejection of the null hypothesis. Also, to strengthen the conclusion regarding H_0 , we will also calculate confidence intervals for our variables.

² PubMed : Trends in adult body-mass index in 200 countries from 1975 to 2014: a pooled analysis of 1698 population-based measurement studies with 19·2 million participants
<https://pubmed.ncbi.nlm.nih.gov/27115820/>

4. Analysis

Our statistical analysis tests 3 hypotheses about factors associated with obesity among Latin American women.

4.1. BMI level compared to global average

We test if the mean BMI of Latin American women is greater than 3 (worldwide average BMI of women based on the previous determined scale).

Hypotheses:

$$H_0: E(imc_level) = 3$$

$$H_1 : E(imc_level) > 3$$

Using our sample, we calculate the value of our estimator which approximates the one for the population:

$$E(imc_level) \approx E(\widehat{imc_level}) = 4.3$$

Our sample reveals that the average BMI level recorded is 4.3, a value that is noticeably higher than the reference value of 3, which was used for the null hypothesis. To put it another way, the mean BMI of Latin American women in our sample is 1.3 units higher than the worldwide standard. The next step is to find out if this difference we have observed is significant enough to discard the null hypothesis and we use the central limit theorem for that.

Using R, we calculate the variability of our estimator as follows:

$$E(\widehat{imc_level}) \sim \mathcal{N}(3, \frac{Var(imc_level)}{n} = 2.1137)$$

We have to test if this variability allows the rejection of the null hypothesis and for this, we calculate the p value.

We start by calculating the z-score of our test, obtaining 20.18 . This z-score allows us to calculate the p-value of our test, which is 10^{-90} being strictly less than 0.05, we can reject H_0 .

Another way of finding out whether we can reject the H_0 hypothesis is by calculating the confidence interval.

$$\begin{aligned} \text{lower_limit} &= 4.3212 - 1.96 \cdot 0.0654 = 4.26 \\ \text{upper_limit} &= 4.3212 + 1.96 \cdot 0.0654 = 4.38 \end{aligned}$$

So, the confidence interval at 95% [4.26,4.38]. Since this interval does not contain the value 3, this confirms the rejection of H_0 .

To conclude this analysis, based on the results, we say that the BMI level of Latin American women is significantly higher (This falls within the obesity range of the BMI scale) than the worldwide average BMI level for women. This shows that women in Latin America could be more likely affected by obesity than those in other regions.

Now that we have observed a difference in BMI levels between Latin America and the rest of the world, we can examine the other two variables to determine whether they influence the risk of obesity.

4.2. Relation between the BMI level (imc_level) and age of the women:

We test if mean **age** differs between obese and non-obese women. For that, to separate BMI level in 2 sub-groups for our analysis, we use our variable **obesity_level**.

Hypotheses:

$$H_0: E(\text{age} | \text{obesity_level} = 1) - E(\text{age} | \text{obesity_level} = 0) = 0$$

$$H_1: E(\text{age} | \text{obesity_level} = 1) - E(\text{age} | \text{obesity_level} = 0) \neq 0$$

With our sample we find the following value:

- n_1 represents the number of non obese women ($n_1=325$)
- n_2 represents the number of obese women ($n_2 = 718$)
- $n = n_1 + n_2$ is the total size of our sample ($n = 1043$)

With our sample, we calculate the value of the estimated difference from our sample as in our hypothesis which approximate the value for the population:

$$\begin{aligned} E(\text{age} | \text{obesity_level} = 1) - E(\text{age} | \text{obesity_level} = 0) &\approx \\ E(\widehat{\text{age} | \text{obesity_level} = 1}) - E(\widehat{\text{age} | \text{obesity_level} = 0}) &= 3.80 \end{aligned}$$

We can see that there is a difference of 3.80 between the age of the sub sample. It means that based on our sample, the women having obesity are about 4 years older than the non-obese women. We then need to determine if the difference between the value associated with our null hypothesis and the estimator allows the rejection of H_0 . To do so, we analyze the degree of significance of the variability of the estimator, we use the Central Limit Theorem.

With R, we calculate the variability of our estimator as follows:

$$\sim \mathcal{N} \left(0, \frac{E(\widehat{\text{age} | \text{obesity_level} = 1}) - E(\widehat{\text{age} | \text{obesity_level} = 0})}{n_1} + \frac{E(\widehat{\text{age} | \text{obesity_level} = 1}) - E(\widehat{\text{age} | \text{obesity_level} = 0})}{n_2} \right) = 0.3685$$

Using p-value, we need to see if the variability of the estimator indicates a rejection of our hypothesis.

We calculate our z-score, and it was 10.316 with R then it allows us to calculate the p-value which was around 0.0000001 so the p-value is strictly less than 0.05 so we can reject H_0 .

Another method to find out the validity of the H_0 is with the confidence interval that we calculate below.

$$\text{lower_limit} = 3.80 - 1.96 \cdot 0.3685 = 3.08$$

$$\text{upper_limit} = 3.80 + 1.96 \cdot 0.3685 = 4.52$$

So, our confidence interval at 95% is [3.08, 4.52]. We can see that this interval does not contain 0, which confirms the test result. At the 5% threshold, we rejected H_0 .

In conclusion, the subgroup living with obesity is older than the subgroup without obesity. These results may suggest that aging could be a factor associated with obesity in our population.

4.3. Relation between the BMI level (imc_level) and physical activity (pal)

We test whether obese women have a lower mean frequency of physical activity (**pal**). For that, we also separate BMI levels in 2 subgroups for our analysis using our variable **obesity_level**.

Hypotheses:

$$H_0: E(pal|obesity_level = 1) - E(pal|obesity_level = 0) = 0$$

$$H_1: E(pal|obesity_level = 1) - E(pal|obesity_level = 0) < 0$$

With our sample, we calculate the value of the estimated difference from our sample as in our hypothesis:

$$\begin{aligned} E(pal | obesity_level = 1) - E(pal | obesity_level = 0) &\approx \\ E(pal|\widehat{obesity_level} = 1) - E(pal|\widehat{obesity_level} = 0) &= -0.350 \end{aligned}$$

We observe a negative difference, suggesting that obese women have a lower physical activity level. Then we need to determine if this difference between our null hypothesis and the estimator allows his rejection. To do so, we analyze the degree of significance of the variability of the estimator, we use the Central Limit Theorem.

Using R, we calculate the variability of our estimator as follows:

$$\begin{aligned} &E(pal|\widehat{obesity_level} = 1) - E(pal|\widehat{obesity_level} = 0) \\ \sim \mathcal{N} \left(0, \frac{Var(pal|obesity_level = 1)}{n_1} + \frac{Var(pal|obesity_level = 0)}{n_2} \right) &= 0.0034619 \end{aligned}$$

We start by calculating the z-score of our test, obtaining -5.94 this z-score allows us to calculate the p-value of our test, which is 1.36×10^{-9} being strictly less than 0.05, we can reject H_0 .

Another way of finding out whether we can reject the H_0 hypothesis is by calculating the confidence interval.

$$\begin{aligned} \text{lower_limit} &= -0.350 - 1.96 \cdot 0.05887 = -0.465 \\ \text{upper_limit} &= -0.350 + 1.96 \cdot 0.05887 = -0.235 \end{aligned}$$

Our confidence intervals at 95% is [-0.465, -0.235]. Since this interval does not contain the value 0, this confirms the rejection of H_0 .

We can conclude this part by saying that Obese women have a lower mean frequency of physical activity than non-obese women. These results suggest that few physical activities could be associated with obesity in our population.

5. Limitations

We were unable to obtain more recent values (e.g. 2025). Our dataset dates from 2019³, thus our result may not fully reflect the current reality.

For the variable **pal**, we had to interpret the meaning of its numerical scale, as the data source did not provide detailed explanations for the variable's coding.

Some variables in our dataset, such as **pal**, touch upon sensitive or personal topics. As a result, participants may not have responded fully or accurately, which could introduce bias into our findings.

For Hypothesis 1, the results showed that the mean BMI in our sample is greater than the world average, indicating a prevalence of obesity among Latin American women. However, this does not allow us to determine the exact factors causing this higher average. Which can be more than just age and physical activity. It remains a valuable result which emphasizes the specific situation of the Latin American population.

We also set the value $\mu_1 = 3$ for this hypothesis based on a global female BMI reference score of 24,4 from 2016⁴. We were unable to locate a more recent reference value. Consequently, the results of this test rely on a benchmark that may no longer be fully valid in 2025. Moreover, this reference is older than the sample data itself (2019), which limits the timeliness of the comparison.

Another limitation of our analysis is that we treat the two groups (obese and non-obese women) as if they were completely independent and similar within each group. However, we do not consider other important factors, such as socioeconomic status or living environment. These missing factors could help explain some of the differences we observe between the two groups.

³ Palechor, F. M., & Manotas, A. D. L. H. (2019). Dataset for estimation of obesity levels based on eating habits and physical condition in individuals from Colombia, Peru and Mexico. *Data in Brief*, 25, Article 104344. <https://doi.org/10.1016/j.dib.2019.104344>

⁴ PubMed : Trends in adult body-mass index in 200 countries from 1975 to 2014: a pooled analysis of 1698 population-based measurement studies with 19·2 million participants
<https://pubmed.ncbi.nlm.nih.gov/27115820/>

6. Discussion

Through this study and based on our sample, we gained a better understanding of the situation of Latin American women regarding obesity and examined the relationship between BMI levels, age, and physical activity.

Our analysis reveals, first, that Latin American women are at a higher risk of obesity compared to the global average. Second, we found significant differences in the explanatory variables between non-obese and obese women. This demonstrates a significant association between obesity and two key factors: age and physical activity level.

The elevated average BMI level (4.32) compared to the global average (3.0) underscores the critical obesity situation in Latin America. This finding aligns with the information mentioned in the introduction which indicates a high prevalence of obesity in the region.

Our results show that women living with obesity are approximately 3.8 years older than those without, confirming that aging could be a factor associated with obesity in Latin America. These findings align with the biological understanding of age-related metabolic changes; as women age, metabolic rate typically declines.

Furthermore, the analysis shows significantly lower physical activity levels among women living with obesity. It highlights the impact of sedentary behavior and the importance of energy expenditure in weight management.

These findings have implications for public health strategies in Latin America. Interventions should be multifaceted: promoting

physical activity while considering age-specific approaches and investigating other potential factors not covered in this study.

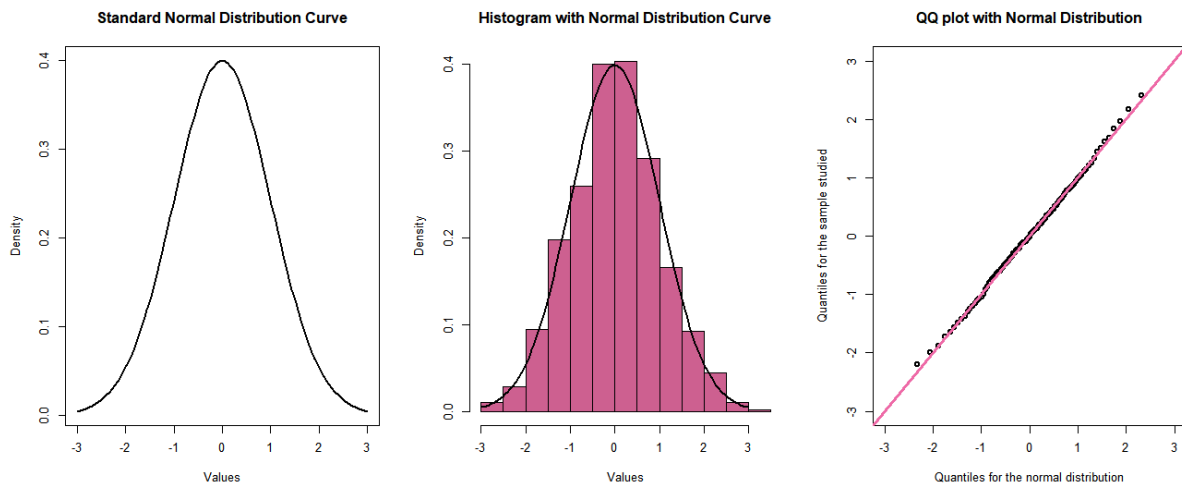
Future research should address the limitations of our work by including additional variables and collecting more current data. Despite these limitations, our study provides strong evidence supporting the importance of age and physical activity in understanding obesity risk among Latin American women.

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8. Appendix – Supplements

A. Appendix - Normal model



B. Appendix – R Code

#creation our database - We import a database from Kaggle but only with women filtered with Excel

```
setwd("C:/Users/MSI/Documents/3eme semestre bachelor/business statistics 2/projet")
```

```
getwd()
```

```
women <- read.csv("DataBaseWomen.csv", header = TRUE, sep = ";")
```

#selection of variables from the original database

```
women2 <- women[, c("Age", "Weight", "Height", "FAF", "IMC")]
```

#change variable names

```
names(women2) <- c("age", "weight", "height", "pal", "imc")
```

#definition of the BMI level scale

```
women2$imc_level <- ifelse(women2$imc < 16, 0,
```

```
  ifelse(women2$imc < 19, 1,
```

```
  ifelse(women2$imc < 21, 2,
```

```
  ifelse(women2$imc < 25, 3,
```

```
  ifelse(women2$imc < 30, 4,
```

```
  ifelse(women2$imc < 35, 5,
```

```
  ifelse(women2$imc < 40, 6, 7))))))
```

#define whether a woman is obese or not based on the imc_level

```
women2$obesity_level <- ifelse(women2$imc_level < 4, 0, 1)
```

#correction of the database with commas and integers

```

# age -> integers
women2$age <- as.integer(as.numeric(sub(",", ".", women2$age, fixed = TRUE)))

# height, weight, pal and imc have a comma, we replace it with a dot before converting
women2$height <- as.numeric(sub(",", ".", women2$height, fixed = TRUE))
women2$weight <- as.numeric(sub(",", ".", women2$weight, fixed = TRUE))
women2$pal <- as.numeric(sub(",", ".", women2$pal, fixed = TRUE))
women2$imc <- as.numeric(sub(",", ".", women2$imc, fixed = TRUE))
View(women2)

# To extract the database we have created
write.csv2(women2, "DataBaseDentBlanche.csv", row.names = FALSE)

#-----

#Code for the Central Limit Theorem

#First, we need a vector containing our variable imc_level
imc_level_v <- women2$imc_level

#The proportion of each obesity level in the sample
(sum(imc_level_v == 0)/length(imc_level_v)) *100
(sum(imc_level_v == 1)/length(imc_level_v)) *100
(sum(imc_level_v == 2)/length(imc_level_v)) *100
(sum(imc_level_v == 3)/length(imc_level_v)) *100
(sum(imc_level_v == 4)/length(imc_level_v)) *100
(sum(imc_level_v == 5)/length(imc_level_v)) *100
(sum(imc_level_v == 6)/length(imc_level_v)) *100
(sum(imc_level_v == 7)/length(imc_level_v)) *100

#The proportion for each obesity level
df <- as.data.frame(prop.table(table(imc_level_v)) * 100)
colnames(df) <- c("BMI_Level", "Percentage")
df
df <- df[order(-df$Percentage), ] # Now shown in decreasing order
df

#here we show the proportion of each imc_level
props <- table(imc_level_v) / length(imc_level_v) * 100

```

```

pie(props,
     main = "BMI level proportions",
     col = rainbow(length(props)),
     labels = paste0(names(props), " (", round(props, 1), "%)")

#Then we calculate the simple samples for different sizes

#The function
sample_size = function(data, size)
{ replicate(700, mean(sample(data, size)))}

mean_5 = sample_size(imc_level_v, 5)
mean_20 = sample_size(imc_level_v, 20)
mean_100 = sample_size(imc_level_v, 100)

#Display the distribution for obesity levels

#The function
hist_obesity_level <- function(data, main="QQ-Plot", xlab="x value", ylab="frequency")
{
  hist(data, main = main,xlab = xlab,ylab=ylab, xlim=c(0, 7), ylim=c(0,300), breaks = 7, col =
"hotpink3", border = "white")
  abline(v = mean(data), col = "gray1", lwd = 2)
}

#Display the distribution for different sample sizes
par(mfrow = c(1,3))
hist_obesity_level(mean_5,main="Distribution for the sample size of 5", xlab="sample size of 5")
hist_obesity_level(mean_20,main="Distribution for the sample size of 20", xlab="sample size of 20")
hist_obesity_level(mean_100,main="Distribution for the sample size of 100", xlab="sample size of
100")

#Display their Quantile plots

#The function
qqnorm <- function(x, main="QQ-Plot", xlab="Quantiles for the normal distribution",
                  ylab="Quantiles for the sample studied") {
  prob <- seq(0.01, 0.99, 0.01)

```

```

standar <- scale(x)
plot(qnorm(prob), quantile(standar, prob),
     xlim=c(-3,3), ylim=c(-3,3),
     main=main, xlab=xlab, ylab=ylab, pch = 1, lwd = 2)
abline(0,1,col= "hotpink2",lwd=2.5)
}
par(mfrow = c(1,3))
qqnorm(mean_5, main = "qqplot - sample of size 5")
qqnorm(mean_20, main = "qqplot - sample of size 20")
qqnorm(mean_100, main = "qqplot - sample of size 100")

#Check the percentage of the population within different standard deviations from the mean
prop_for_diff_sd = function(data, x)
{
  length(data[data <= mean(data) + x * sd(data) & data >= mean(data) -
             x * sd(data)])/length(data)
}

#For the original population
prop_for_diff_sd(imc_level_v, 1)
prop_for_diff_sd(imc_level_v, 2)
prop_for_diff_sd(imc_level_v, 3)

#For mean_5
prop_for_diff_sd(mean_5, 1)
prop_for_diff_sd(mean_5, 2)
prop_for_diff_sd(mean_5, 3)

#For mean_20
prop_for_diff_sd(mean_20, 1)
prop_for_diff_sd(mean_20, 2)
prop_for_diff_sd(mean_20, 3)

#For mean_100
prop_for_diff_sd(mean_100, 1)
prop_for_diff_sd(mean_100, 2)

```

```
prop_for_diff_sd(mean_100, 3)
```

```
# Optional version for histograms:
```

```
# Adds the total count on each bar (to use this version, just add "2" to the function calls above).
```

```
# Note: some numbers may be slightly misaligned in the last plot.
```

```
hist_obesity_level2 <- function(data,  
                                main = "QQ-Plot",  
                                xlab = "x value",  
                                ylab = "frequency") {  
  h <- hist(data, main = main, xlab = xlab, ylab = ylab,  
            xlim = c(0, 7), ylim = c(0, 300), breaks = 7,  
            col = "hotpink3", border = "white", labels = FALSE)  
  abline(v = mean(data), col = "gray1", lwd = 2)  
  text(h$mids, h$counts, labels = h$counts, pos = 3,  
       col = "gray35", cex = 1, font = 2)}
```

```
#-----
```

```
#code for the description of the data
```

```
#we extract the variables we need
```

```
imc_lvl<-women2$imc_level
```

```
age<-women2$age
```

```
pal<-women2$pal
```

```
obesity_lvl<-women2$obesity_level
```

```
# Get descriptive statistics for each variable
```

```
summary(age)
```

```
summary(imc_lvl)
```

```
summary(pal)
```

```
summary(obesity_lvl)
```

```
# Visualize the distribution of these variables
```

```
hist_var <- function(data, main = "Histogram", xlab = "x value", ylab = "Frequency",breaks=10) {
```

```

data <- na.omit(data)

hist(data,
      main = main,
      xlab = xlab,
      ylab = ylab,
      xlim = range(data),
      breaks = breaks,
      col = "hotpink3",
      border = "black")

abline(v = mean(data), col = "gray1", lwd = 2)
}

par(mfrow = c(1, 2))
hist_var(age,breaks = 10,main="Histogram of age ")
hist_var(pal,breaks = 15,main="Histogram of pal ")
par(mfrow = c(1, 2))

# Create a frequency table for the categorical variable 'imc_level' and 'obesity_level'
tab_imc_lvl <- table(women2$imc_level)
tab_obesity_lvl <- table(women2$obesity_level)

#barplot for the variable "imc_level" and 'obesity_level'
barplot(tab_imc_lvl,
        main = "barplot BMI level",
        xlab = "BMI level",
        ylab = "frequency",
        col = "hotpink3",
        border = "black")

barplot(tab_obesity_lvl,
        main = "barplot obesity level",
        xlab = "obesity level",
        ylab = "frequency",

```

```

col = "hotpink3",
border = "black")

#the qqplot function
my.qqnorm = function(x,main=NULL) {
  scaled = scale(x, center = mean(x))
  sequence = seq(0.01, 0.99, 0.01)
  normal = qnorm(sequence, 0, 1)
  reel = quantile(scaled, sequence)
  variablename = deparse(substitute(x))

  plot(normal, reel,
        xlim = c(-3, 3),
        ylim = c(-3, 3),
        main = main,
        xlab = "Theoretical quantiles",
        ylab = "Sample quantiles",
        pch = 16,
        col = "hotpink3")

  abline(0, 1, col = "gray1", lwd = 2) # similar style to mean line on histogram}
par(mfrow = c(1, 3))

#visualise the QQ-plots of the main variables
my.qqnorm(imc_lvl,main = "QQplot: imc_level")
my.qqnorm(age, main="QQplot:age")
my.qqnorm( pal,main="QQplot:pal")

# Visualise the reference plots we would expect if our variables followed a normal distribution
par(mfrow = c(1, 3))
set.seed(123)
x <- rnorm(1000, 0, 1)
curve(dnorm(x, mean = 0, sd = 1),
      from = -3, to = 3,

```

```

      xlab = "Values", ylab = "Density",
      main = "Standard Normal Distribution Curve",
      lwd = 2)
hist(x,
      breaks = 10,
      freq = FALSE,
      xlab = "Values",
      ylab = "Density",
      main = "Histogram with Normal Distribution Curve",
      col = "hotpink3",
      border = "black")

curve(dnorm(x, mean = 0, sd = 1),
      from = -3, to = 3,
      xlab = "Values", ylab = "Density",
      main = "Standard Normal Distribution Curve", add=TRUE,
      lwd = 2)

qqnorm(x,
      main = "QQ plot with Normal Distribution")

#-----
#Code for analysis of our variable
# analyse hypothesis 1
# -> Compare the mean age between non-obese women (obesity_level = 0) and obese women
#(obesity_level = 1)
nop_age <- women2$age[ women2$obesity_level == 0 ]
estim_mnop1 <- mean(nop_age, na.rm = TRUE) #The estimation of the mean age of non-obese
women
estim_mnop1
op_age<-women2$age[women2$obesity_level == 1]
estim_op1 <- mean(op_age, na.rm = TRUE) #The estimation of the mean age of obese women
estim_op1

```

```

# sample variance and sample size for the non-obese group
s1=var(nop_age)
n1=length(nop_age)
s1/n1          # estimated variance of the sample mean (non-obese)

# sample variance and sample size for obese group
s2=var(op_age)
n2=length(op_age)
s2/n2          # estimated variance of the sample mean (obese)

# variance of the estimator (difference of two independent means)
var_estimeur1=(s2/n2)+(s1/n1)
var_estimeur1

# standard error of the difference of means
se_estimeur1 = sqrt(var_estimeur1)
se_estimeur1

# z-statistic for testing H0: (mean_obese - mean_non_obese) = 0
z1 = ((estim_op1 - estim_mnop1) - 0)/se_estimeur1
z1

# two-sided p-value associated with z1
p1=2*(1-(abs(pnorm(z1))))
p1

# 95% confidence interval for the difference of means (obese - non-obese)
ic95_sup1=((estim_op1 - estim_mnop1) - 0)+1.96 * (se_estimeur1)
ic95_sup1
ic95_inf1=((estim_op1 - estim_mnop1) - 0)-1.96 * (se_estimeur1)
ic95_inf1

# analyse hypothesis 2
# -> Compare the mean pal (physical activity level) between non-obese and obese women
nop_pal <- women2$pal[ women2$obesity_level == 0 ]
estim_mnop2 <- mean(nop_pal, na.rm = TRUE)#The estimation of the mean pal of non-obese women

```

```

estim_mnop2
op_pal<-women2$pal[women2$obesity_level == 1]
estim_op2 <- mean(op_pal, na.rm = TRUE)#The estimation of the mean pal of obese women
estim_op2
# sample variance, standard deviation and sample size for the non-obese group (pal)
s3=var(nop_pal)
sd3=sqrt(s3)
sd3
n3=length(nop_pal)
s3/n3          # variance of the sample mean (non-obese, pal)
# sample variance, standard deviation and sample size for the obese group (pal)
s4=var(op_pal)
sd4=sqrt(s4)
sd4
n4=length(op_pal)
s4/n4          # variance of the sample mean (obese, pal)
# variance and standard error of the difference of mean pal
var_estimeur2=(s4/n4)+(s3/n3)
var_estimeur2
se_estimeur2 = sqrt(var_estimeur2)
se_estimeur2
# z-statistic for H0: difference of mean pal = 0
z2 = ((estim_op2 - estim_mnop2) - 0)/se_estimeur2
z2
# one-sided p-value associated with z2 (direction depends on H1)
p2=pnorm(z2)
p2
# 95% confidence interval for the difference of mean pal
ic95_sup2=((estim_op2 - estim_mnop2) - 0)+1.96 * (se_estimeur2)
ic95_sup2
ic95_inf2=((estim_op2 - estim_mnop2) - 0)-1.96 * (se_estimeur2)

```

```
ic95_inf2
```

```
#analyse hypothesis 3
```

```
# -> One-sample test: check if the mean IMC level of the whole sample is equal to  $\mu_0 = 3$ 
```

```
wp_imc_level <- women2$imc_level
```

```
estim_wp_imc_level <- mean(wp_imc_level, na.rm = TRUE) #The estimation of the mean imc_level of  
the whole female population
```

```
estim_wp_imc_level
```

```
mu0=3          # hypothesized mean IMC level
```

```
# sample variance, standard deviation and size for imc_level
```

```
s5=var(wp_imc_level)
```

```
sd5=sqrt(s5)
```

```
sd5
```

```
n5=length(wp_imc_level)
```

```
n5
```

```
# z-statistic for  $H_0$ : mean imc_level = 3
```

```
z3 = (estim_wp_imc_level-mu0)/(sd5/sqrt(n5))
```

```
z3
```

```
# one-sided p-value for this z-statistic
```

```
p3=1-pnorm(z3)
```

```
p3
```

```
# 95% confidence interval for the mean imc_level
```

```
ic95_sup3=((estim_wp_imc_level) - 0)+1.96 * (sd5/sqrt(n5))
```

```
ic95_sup3
```

```
ic95_inf3=((estim_wp_imc_level) - 0)-1.96 * (sd5/sqrt(n5))
```

```
ic95_inf3
```

C. Appendix -Interpretation of the IMC calculated to obtain our 8 categories based on our source

BMI (Body Mass Index)	Classification according to the WHO
< 16	Severe Thinness
<19	Underweight
19 to 21	Mild Thinness
21 to 25	Normal Weigh
25 to 30	Pre-obesity
30 to 35	Moderate Obesity (Class I)
35 to 40	Severe Obesity (Class II)
> 40	Morbid Obesity (Class III)

D. Appendix -a view of our database

	age	weight	height	pal	imc	imc_level	obesity_level
1	21	64.00000	1.620000	0.000000	24.38653	3	0
2	21	56.00000	1.520000	3.000000	24.23823	3	0
3	23	55.00000	1.500000	1.000000	24.44444	3	0
4	21	80.00000	1.720000	2.000000	27.04164	4	1
5	22	66.00000	1.700000	2.000000	22.83737	3	0
6	29	78.00000	1.530000	0.000000	33.32052	5	1
7	30	82.00000	1.710000	0.000000	28.04282	4	1
8	23	70.00000	1.650000	0.000000	25.71166	4	1
9	52	87.00000	1.690000	0.000000	30.46112	5	1
10	22	60.00000	1.650000	1.000000	22.03857	3	0
11	22	82.00000	1.600000	0.000000	32.03125	5	1
12	23	52.00000	1.600000	2.000000	20.31250	2	0
13	31	68.00000	1.580000	1.000000	27.23922	4	1
14	24	76.00000	1.770000	1.000000	24.25867	3	0
15	21	65.00000	1.500000	2.000000	28.88889	4	1
16	22	49.00000	1.560000	2.000000	20.13478	2	0
17	21	48.00000	1.600000	1.000000	18.75000	1	0
18	21	88.00000	1.750000	3.000000	28.73469	4	1
19	21	75.00000	1.670000	1.000000	26.89232	4	1
20	21	64.00000	1.660000	0.000000	23.22543	3	0
21	21	65.00000	1.530000	0.000000	27.76710	4	1
22	20	60.00000	1.660000	1.000000	21.77384	3	0
23	21	50.00000	1.550000	0.000000	20.81165	2	0
24	21	54.50000	1.610000	0.000000	21.02542	3	0
25	20	44.00000	1.500000	0.000000	19.55556	2	0

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Neuchâtel, Décembre 2021

Les auteurs.

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